1. Restore natural knee function

Empowering Kinematic Restoration of the Knee to restore natural Knee function

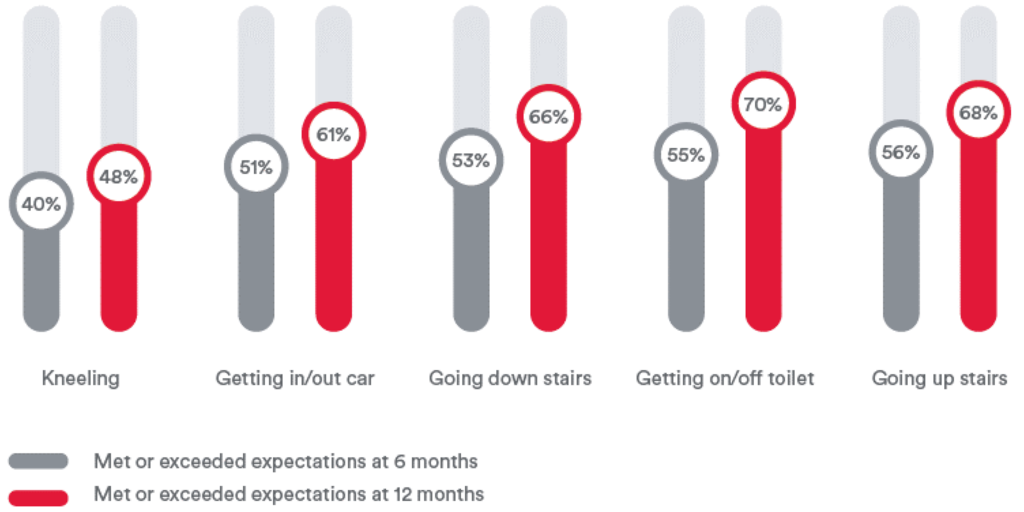
Johnson&Johnson
MedTech

The Challenge

Help you to do better for your patients.

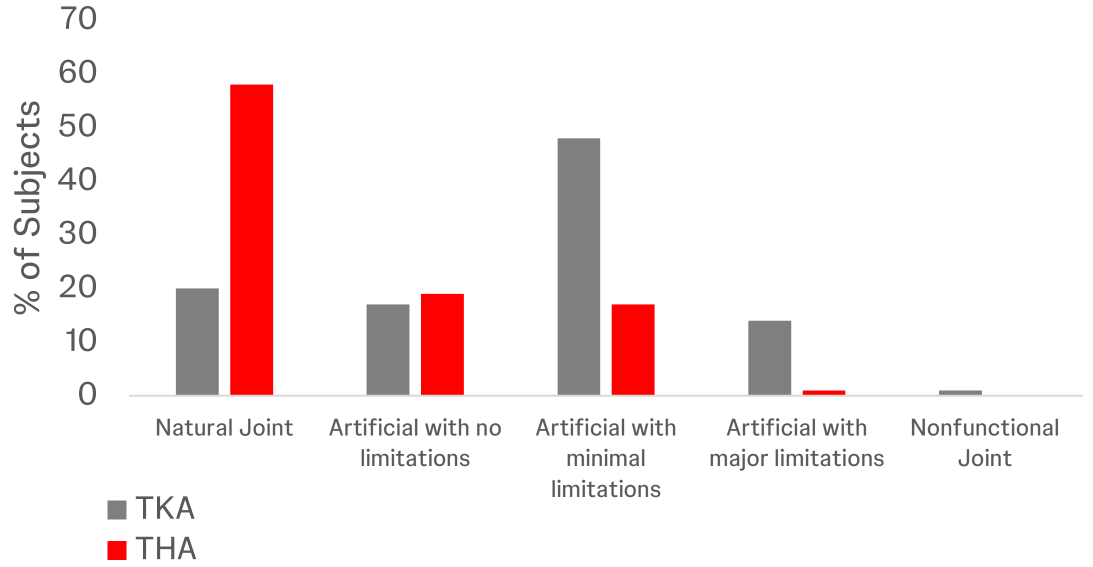
At 6 and 12 months postoperative, ~30% to 60% of patients report limitations in meeting functional expectations, like getting on/off a toilet or walking up/down stairs[1] *

Pre-op Expectations Met/Exceeds at 6 & 12 months Post-op[1]



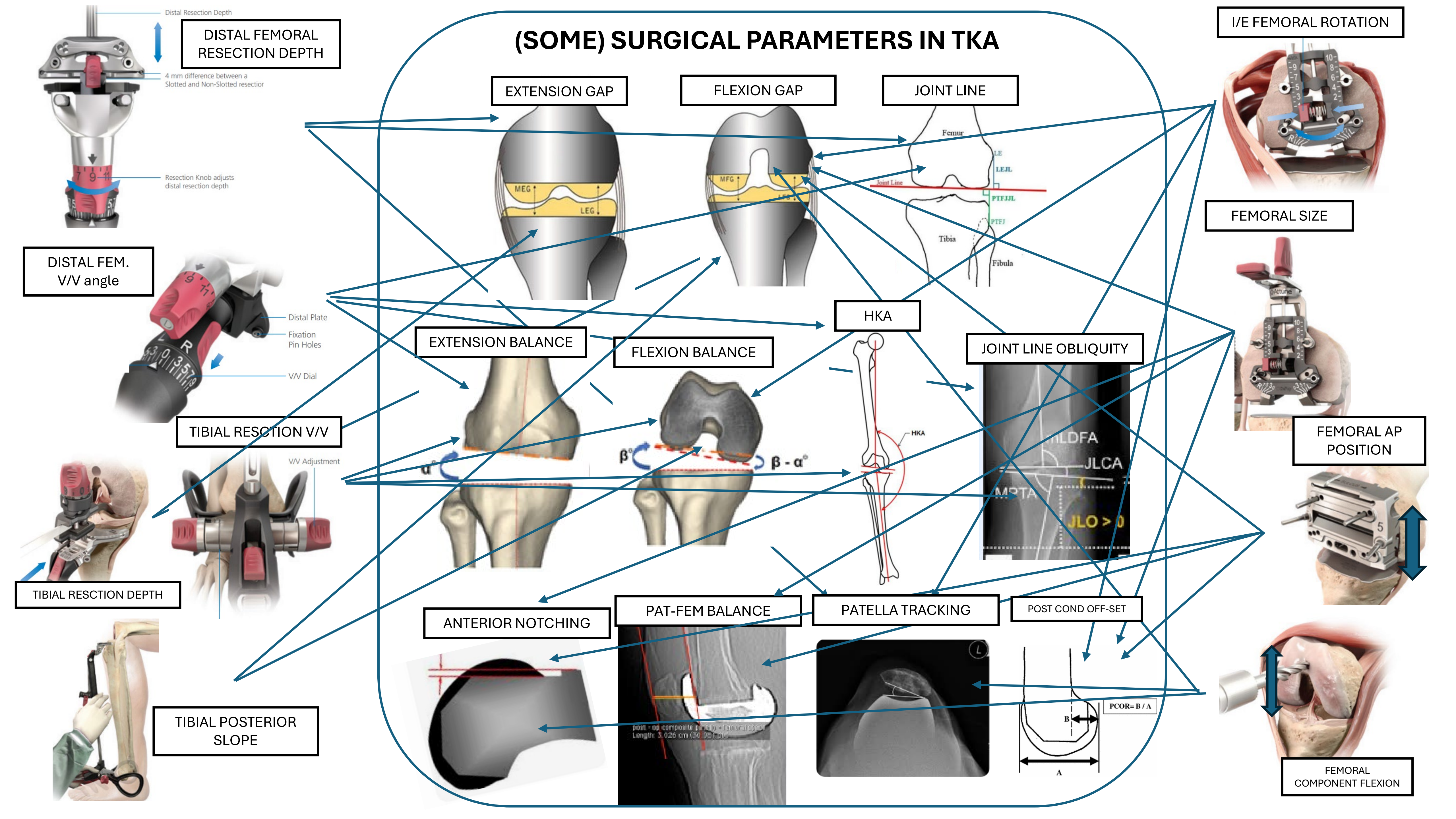
80% of TKA patients do not experience a natural joint vs. 42% of THA patients[2] **

Patient Perception of TKA and THA[2]



* Study conducted in USA

** Study conducted in Canada



(SOME) SURGICAL PARAMETERS IN TKA
Distal Resection Depth
DISTAL FEMORAL RESECTION DEPTH
4 mm difference between a Slotted and Non-Slotted resector
Resection Knob adjusts distal resection depth
DISTAL FEM. V/V angle
Distal Plate
Fixation Pin Holes
V/V Dial
TIBIAL RESCTION V/V
V/V Adjustment
TIBIAL RESCTION DEPTH
TIBIAL POSTERIOR SLOPE
EXTENSION GAP
MEG
LEG
FLEXION GAP
MFG
LFG
JOINT LINE
Femur
LE
LEJL
Joint Line
PTFJJL
PTFJ
Tibia
Fibula
HKA
EXTENSION BALANCE
FLEXION BALANCE
α°
β°
β - α°
JOINT LINE OBLIQUITY
mLDFA
JLCA
MPTA
JLO > 0
ANTERIOR NOTCHING
PAT-FEM BALANCE
PATELLA TRACKING
POST COND OFF-SET
PCOR= B / A
B
A
I/E FEMORAL ROTATION
FEMORAL SIZE
FEMORAL AP POSITION
FEMORAL COMPONENT FLEXION

Kinematic Restoration

Kinematic Restoration is the art and science of restoring knee function.

- From pre-op through post-op, from incision to closure—it provides a **holistic approach** that works in harmony with kinematic and inverse kinematic alignment philosophies
- The VELYS™ Robotic-Assisted Solution equips surgeons with **real-time** insights to optimize implant placement, preserve soft tissue envelope, predict joint stability and work toward returning knee function.[5-7,42]
- ATTUNE™ Knee is verified to work with kinematic alignment techniques within tested boundaries.[3-6*]

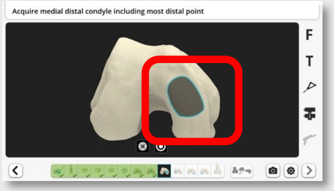
The powerful combination can reduce soft tissue releases to improve function and may accelerate patient recovery.[7]

*Based on in vitro studies.

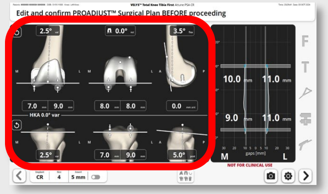
Johnson & Johnson MedTech Knee Platform Solutions

Kinematic Restoration

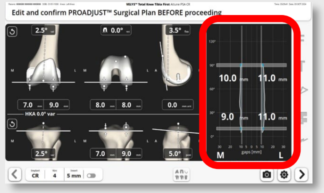
Surgeon driven landmark acquisition



Surgeons control the 3D surgical plan (e.g. rKA, iKA, MA)

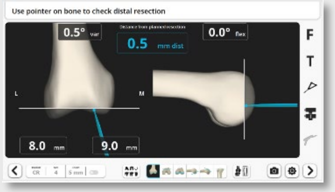


Pre-resection soft tissue balance check



Surgeon controlled, Robotic-Assisted resection

Immediate accuracy verification & ability to adjust the plan



Implantation of Kinematically-advanced ATTUNE Knee

Stability

Multiple studies have demonstrated **ATTUNE to exhibit smooth roll-back without abrupt change at mid-flexion**[8-11]

Satisfaction

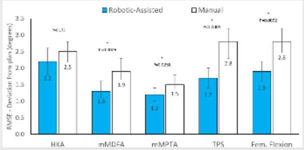
ATTUNE Knee is delivering better patient satisfaction than the class of other knees.[12¶]

90%
Of patients are feeling better following TKA with ATTUNE Knee.[1§]

88%
Of patients are satisfied with the results of their ATTUNE Knee TKA surgery.[2¶]

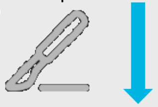
Improved Accuracy

Studies demonstrate VELYS RAS has improved implant alignment accuracy compared to Manual.[13,14 †]



Fewer Soft-Tissue Releases

The VELYS Robotic-Assisted Solution showed significantly lower incidence of intentional soft tissue release compared to manual TKA.[7]



P<0.0001

Improved Patient Function & Outcomes

After one year of follow-up, patients treated with the VELYS™ Robotic-Assisted Solution showed statistically significant improvements ($p<0.05$) in their Oxford Knee Score, WOMAC, Forgotten Knee Score, KOOS ADL, Activity Pain, and Range of Motion compared to those treated with the BRAINLAB KNEE3 Navigation System[15*]

P<0.05
Oxford, WOMAC, KOOS, FKS
vs KNEE3 Navigation

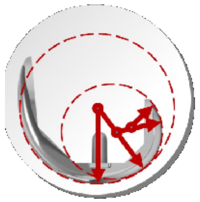
¶ According to the NJR ISR for ATTUNE Cemented, 88% of patients reported satisfaction in responding to the question "How would you describe the results of your operation?
† Product may not be commercially available in all markets. This content may not be used externally in those markets where regulatory approval has not been granted for all the products referenced.
* In a prospective study of 90 TKA patients.
§ According to a Patient Satisfaction PROMS analysis 90% reported improvement in responding to the question "Overall, how are your problems now, compared to before your operation".

Limitations

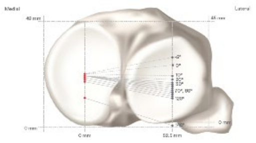
How can you do better for your patients?

Implant

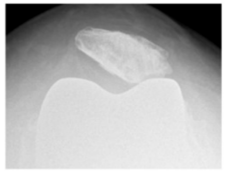
1. Research has shown that legacy Designs can contribute to mid-flexion Instability which can lead to a lack of patient confidence in the knee.[16]



2. The natural knee is seen to be very stable medially and roll back laterally, medial instability has been associated with worse outcomes.[17]

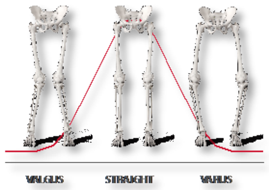


3. Patella Femoral tracking can be compromised when performing patient specific techniques.[18]

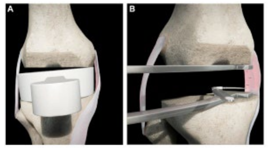


Technique

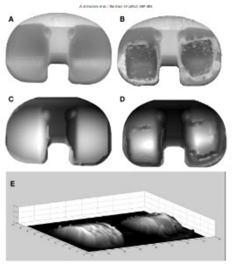
1. Mechanical alignment is the current gold standard tech, but **only 0.1% of patients have neutral femurs and tibias.**[19]



2. Mechanical alignment requires soft tissue releases to balance the knee in a high proportion of cases.[20]

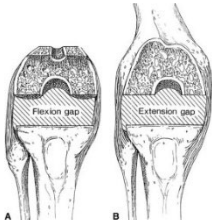


3. Implants designed and tested under mechanical alignment conditions. Very limited published implant testing under alternative alignment conditions which may influence wear and fixation performance.[21]



Technology

1. Manual instruments have reduced accuracy when implementing patient specific techniques and lack predictive data on balancing.[22]

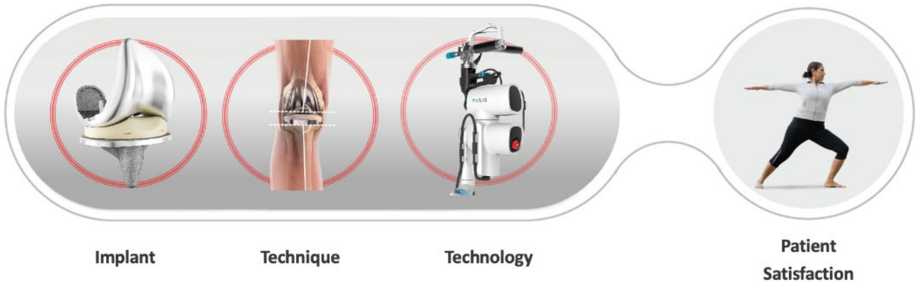


2. Navigation systems require positioning and pinning of cutting blocks adding additional steps to the procedure.[23]

3. CT based Robotic systems are reliant on technician defined landmark acquisition.[24]

J&J Knee Platform Solutions

Clinically empowering Kinematic Restoration of the Knee to restore natural Knee function

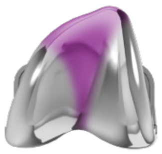


Implant

1. The ATTUNE GRADIUS™ Curve is the gradually reducing femoral radius designed to provide a smooth transition from stability to mobility through a patient's range of motion.[8-11]

2. ATTUNE™ Medial Stabilized Knee System
   - Reduces A/P motion without abrupt direction changes
   - Increases conformity on the medial compartment
   - Lateral compartment is flatter to encourage natural femoral rollback[8,11,25,26]

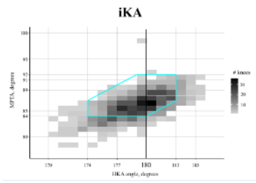
3. ATTUNE™ Knee Advanced Patellofemoral Articulation geometry accommodates variations in trochlear orientation required for kinematic philosophies[27, 43-49]



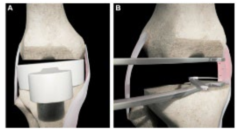
J&J MedTech

Technique

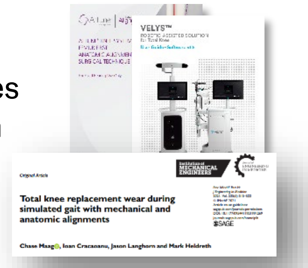
1. Alternative alignment techniques such as rKA and iKA are capable of restoring the alignment and joint line obliquity for a high proportion of the population[28,29]



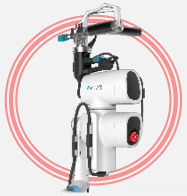
2. Alternative alignment techniques such as rKA and iKA reduce the need for soft tissue releases and adjustments.[28-30]



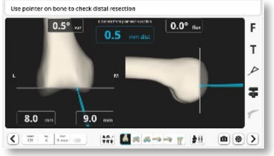
3. ATTUNE has approved techniques supporting alternative alignment techniques and have published evidence on performance of ATTUNE in alternative alignments.[31-33]



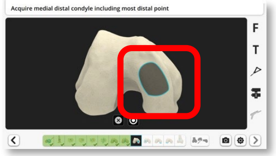
Technology



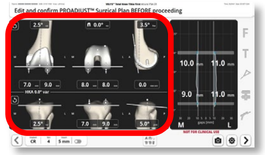
1. Parameters are easily adjusted through the VELYS™ Robotic-Assisted Solution planning screen, helping surgeons to personalize alignment and balance relative to soft tissues.[32]



2. VELYS RAS maintains the saw cut plane to help execute precise, reproducible, surgeon-controlled cuts without cutting blocks[13,14,32]

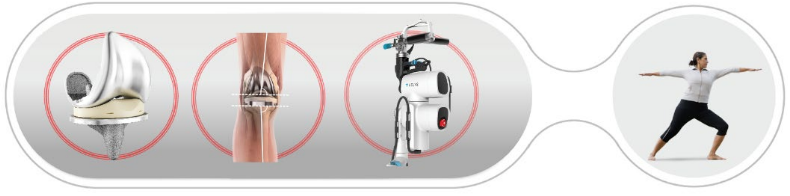


3. Surgeons dictate the selection of the critical points and define the 3D surgical plan[32]



ATTUNE Evidence

Clinically empowering Kinematic Restoration of the Knee to restore natural Knee function



ATTUNE™ Knee System

Implant

1. Multiple studies have demonstrated ATTUNE GRADIUS™ Curve to exhibit smooth roll-back without abrupt change at mid-flexion.[8-11]

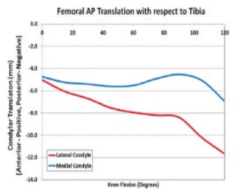
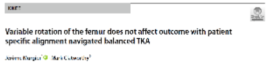
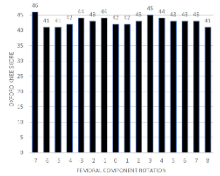


Figure 1. Lateral and Medial Condylar translation during DKB

2. Simulation has shown that the ATTUNE Medial Stabilized Knee System has lower overall medial condyle motion and does not see the abrupt changes in direction associated with implants based on a multi-radius sagittal geometry.[25∞]


Variable rotation of the femur does not affect outcome with patient specific alignment navigated balanced TKA

3. A Study on the ATTUNE Knee with a patient specific technique found no influence of femoral rotation on PROMs and no patella complications, suggesting the patella femoral design is robust to positioning outside mechanical alignment.[34]



4. ATTUNE Knee is delivering better patient satisfaction than the class of other knees.[12*]

90% Of patients are feeling better following TKA with ATTUNE Knee.[1†]

88% Of patients are satisfied with the results of their ATTUNE Knee TKA surgery.[2*]

∞ Shows the results of ATTUNE CR, ATTUNE MS, Persona MC, and Triathlon CS in both CR and CS applications. In CR application, medially, both the ATTUNE CR and MS implant constructs demonstrated minimal Femoral A-P translation, which stayed constant during most of the flexion. Both constructs experienced posterior femoral translation (rollback). The Persona MC experienced similar medial A-P translation as that of the ATTUNE CR and MS. However the A-P direction of translation reversed at about 25 degrees and 60 degrees of knee flexion. CS application, medially the ATTUNE MS construct experienced the least amount of total A-P translation, the ATTUNE MS, Persona MC and Triathlon CS (in that order). Conclusion: Demonstrates differences between the LP motion patterns, ligament force contributions and their strains for all simulated implant constructs in CR and CS application. The CS application, without PCL, the femur could slide anteriorly with flexion if the insert does not provide desired constraint to the anterior translation. Results: Demonstrated the effect of the size mis-matched Femurs in the Persona MC constructs. Trends indicated that the Femur-insert articulation for the smaller Femur size may offer less medial A-P stability that is achieved the the larger sized Femur. The subsets of the results were normalized to their starting point at full extension and compared within designs. *According to the NJR ISR for ATTUNE Cemented, 88% of patients reported satisfaction in responding to the question "How would you describe the results of your operation?" †According to a Patient Satisfaction PROMS analysis 90% reported improvement in responding to the question "Overall, how are your problems now, compared to before your operation"

Kinematic Alignments

Technique

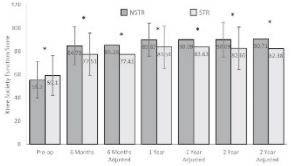
1. iKA utilizing VELYS RAS allows accurate reconstruction of joint line height & obliquity compared to MA.[35]

Table 2 Changes in joint line obliquity and joint line hei

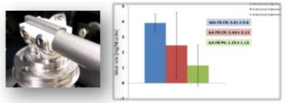
	Pre-op value	Post-op value	*P*-value
MA Group (*n* = *101*)			
Mean JLO, Degrees (SD)	2.94 (1.9)	2.31 (1.5)	0.004
Mean JLH, mm (SD)	40.6 (4.5)	40.6 (4.8)	0.89
iKA Group (*n* = *101*)			
Mean JLO, Degrees (SD)	2.43 (1.8)	2.30 (1.4)	0.57
Mean JLH, mm (SD)	41.2 (6.6)	42.4 (6.2)	0.17

MA, Mechanical Alignment; iKA, Inverse Kinematic Alignment; JLO, Joint Line Obliquity; JLH, Joint Line Height; SD, Standard Deviation

2. Avoiding soft tissue releases when implanting the ATTUNE™ Knee System has been associated with improved functional outcomes.[7]



3. Published in vitro testing on the ATTUNE™ Knee Systems showed no increase in wear and equivalent cement fixation under simulated alternative alignment conditions as compared to mechanical alignment. [33,36**]

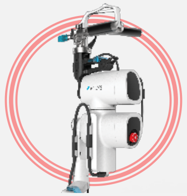


4. The use of iKA and KA have been associated with either equivalent or improved short term clinical and functional outcomes compared to mechanical alignment.[37,38]

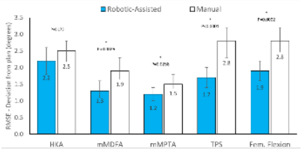
** In vitro testing of components positioned with a simulated HKA of 3° Varus and a tibial joint line of 5° Varus

VELYS™ Robotic-Assisted Solution

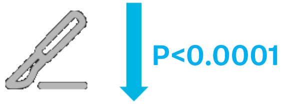
Technology



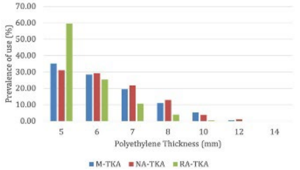
1. Studies demonstrate the use of VELYS™ Robotic-Assisted Solution results in improved implant alignment accuracy compared to manual.[13,14 §]



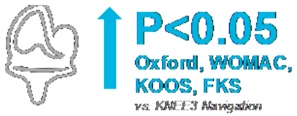
2. The VELYS Robotic-Assisted Solution showed significantly lower incidence of intentional soft tissue release compared to manual TKA.[7]



3. VELYS Robotic-Assisted solution has demonstrated reduced navigation time and improved accuracy compared to Navigation.[39-41]



4. After one year of follow-up, patients treated with the VELYS™ Robotic-Assisted Solution showed statistically significant improvements (p<0.05) in their Oxford Knee Score, WOMAC, Forgotten Knee Score, KOOS ADL, Activity Pain, and Range of Motion compared to those treated with the BRAINLAB KNEE3 Navigation System[15¶]



¶ In a prospective study of 90 TKA patients.
§ Product may not be commercially available in all markets. This content may not be used externally in those markets where regulatory approval has not been granted for all the products referenced.

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